



Republic of Rwanda
Ministry of Education



BDCAC501

ARCHICAD SOFTWARE

Apply ArchiCAD Software

Competence

RQF Level:

5

Learning Hours



Credits:

7

Sector:

Construction and Building Services

Trade:

Building Construction

Module Type:

Specific

Curriculum:

CBSBDC5001-TVET Certificate V in Building Construction

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Purpose statement	This specific module describes the skills, knowledge, and attitudes required to use ArchiCAD software. It is meant for learners who have successfully completed level IV in building construction and pursuing level V in building construction. At the end of this module, learners will be able to prepare the ArchiCAD software environment, produce ArchiCAD drawings, and print out ArchiCAD drawings.					
Learning assumed to be in place	N/A					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative Assessment			50%

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
Prepare ArchiCAD software environment	1.1 ArchiCAD software requirements are properly analyzed based on the operating system
	1.2 ArchiCAD software is properly installed based on the Operating System
	1.3 Software is properly customized as per the installation user guide
Produce ArchiCAD Drawings	2.1 Project units are properly set up according to the scale of the drawing
	2.2 Software tools are Appropriately used according to the design objectives
	2.3 Software 2D drawings are appropriately created according to desired design
	2.4 Software 3D modeling is appropriately created according to desired design
	2.5 Title block is appropriately created according to the standard requirements
Print out ArchiCAD drawings	3.1 Master layout book is appropriately created according to the scale of the drawing
	3.2 Placing Views on the layout is placed on layout in accordance to scale of drawing
	3.3 Project layout is appropriately saved according to the given project

	3.4 Layout books are appropriately printed out according to the required document format.
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Intended Knowledge, Skills and Attitude

Knowledge	Skills	Attitude
✓ Describe architectural design principles and theories ✓ Identify building materials ✓ Describe construction methods and building services ✓ Describe the ArchiCAD software's capabilities, features, and limitations ✓ Identify building codes, regulations, and industry standards ✓ Describe construction documentation ✓ Identify industry trends, emerging technologies, and software updates	✓ Perform ArchiCAD installation ✓ Produce 2D drawings such as plans, sections, and elevations ✓ Produce 3D drawings such as perspectives. ✓ Perform navigation in ArchiCAD software. ✓ Use ArchiCAD design tools ✓ Perform high-quality visualizations and renderings ✓ Integrate ArchiCAD with other design and analysis software ✓ Print out a drawing	✓ Make continuous learning and skill development ✓ Pay attention to detail and precision in design and documentation ✓ Adapt to software updates and industry requirements ✓ Aware of collaborative mindset for effective coordination with project stakeholders ✓ Proactive to problem-solving and troubleshooting abilities ✓ Willing to experiment and explore new design possibilities ✓ Effective time management and organizational skills ✓ Conduct ethical and professional architectural practice

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Prepare ArchiCAD software environment 2. Produce ArchiCAD Drawings 3. Print out ArchiCAD drawings
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Learning outcome 1: Prepare ArchiCAD Software Environment	Learning hours: 10
Indicative content	
• Analysing ArchiCAD software requirements ✓ Description of ArchiCAD: ✚ History ✚ Advantages ✚ Applications ✚ Features ✓ Identification of ArchiCAD requirements ✓ Introduction to Building Information Modelling (BIM) system ✚ History ✚ interoperability ✓ Interacting ArchiCAD and Building information modeling (BIM) system ✚ Create Model ✚ Integrating Data ✚ Collaboration ✚ Detecting Clash ✚ Quantity take-off • Installation of ArchiCAD Software ✓ Run the installation file ✓ Activate your ArchiCAD license ✓ Configure ArchiCAD settings • Customization ArchiCAD Software ✓ Set Up Work Environment ✓ Configuration Project Templates ✓ Library Management ✓ Use shortcut keys	
Resources required for the indicative content	
Equipment	• Computer, projector, Whiteboard
Materials	• flip chart, mark pens, Duster, Chalks, Papers

Tools	• Mouse and mouse mat, Electrical cables, - Storage devices (flash disc, CD, External disk), Software protocols, ArchiCAD software
Facilitation techniques	Group discussions, Simulations, Scenario, Trainer Guided • The trainer asks the students to discuss on the ArchiCAD software requirements and building information modelling. The trainer presents the inter relationship of both ARCHICAD and BIM • The trainer introduces the software installation file to students, demonstrate how to install the ArchiCAD software putting in consideration the software license and settings, and asks the students to follow the procedures. • The trainer shares the software file to students in manageable groups and the students practise the exercise of installing the software • The trainer through the installed software demonstrates how to set up work environment, configure project templates, library management and how to use shortcut keys and create architectural drawings asks the students to follow the procedure. • In manageable groups, the trainer asks the students to discuss on setting up work environment, configure project templates and practise use of shortcuts. • The trainer provides an expert view after each presentation.
Formative assessment methods	• Written assessment • Oral assessment • Practical assessment

Learning outcome 2: Produce ArchiCAD drawings	Learning hours: 50
Indicative content	
• Set up project units ✓ Selecting option menu ✓ Selecting project preferences • Using Software tools ✓ Identification of ArchiCAD tools ✚ Design Tools ✚ Documentation Tools ✚ Visualization Tools ✚ Modelling Tools ✚ Collaboration Tools ✚ Visualization & Rendering ✚ Analytical Tools ✚ Automation & Customization ✚ Library & Object Management ✚ Site Tools	

✓ Applying software tools • Creating 2D drawings ✓ Produce 2D Plan views ✓ Make cross-section views ✓ Make elevation views • Creating 3D modelling ✓ Set 3D projections ✓ Set 3D rendering ✓ Generate 3D perspectives • Creating title block ✓ Select layout book ✓ Create a new master layout ✓ Define paper size ✓ Drawing title block elements	
Resources required for the indicative content	
Equipment	• Computer, projector, Whiteboard
Materials	• flip chart, mark pens, Duster, Chalks, Papers
Tools	• Mouse and mouse mat, Board, Electrical cables, - Storage devices, (flash disc, CD, External disk), Software protocols, and software video tutorials
Facilitation techniques	Group discussions, Simulations, Scenario, Trainer guided • The trainer demonstrates to the students how to set up project units, using software tools and asks the student to follow the procedures • Trainer demonstrate to the students the way applying ArchiCAD software tools used to produce drawings and asks the students to follow the procedure. • The trainer provides practical exercises on analyzing and interpreting produced ArchiCAD softcopy drawing documents. • Trainer forms small and manageable groups of students and asks the respective groups to discuss on ArchiCAD Software tools in the creation of 2D and 3D architectural drawings. • The trainer provides enough time to practice what has been taught by giving students different practices, different reference tutorial videos and asking them to present the 2D and 3D architectural drawings. • After submitting, each and every presentation, the trainer provides an expert view of the presented works
Formative assessment methods	• Written assessment • Oral assessment • Practical assessment

Learning outcome 3: Print out ArchiCAD Drawings		Learning hours: 10
Indicative content		
• Creating Master layout book ✓ Identification of the paper sizes of the master layout ✓ Description of a paper size appropriate for drawing ✓ Adjustment of paper size and scale ✓ Allocating Title block • Placing Views on the layout ✓ Drag and place 2D views ✓ Drag and place 3D views ✓ Adjusting View Settings • Saving Project layout ✓ Select master layout ✓ Select the file menu ✓ Choose file formats • Printing of layout books ✓ Review published or exported PDF drawings ✓ Print drawings ✓ Check hardcopy drawings		
Resources required for the indicative content		
Equipment	- Computer, projector, whiteboard, printer	
Materials	Flip chart, mark pens, duster, chalks, papers, printer ink	
Tools	Mouse and mouse mat, board, electrical cables, - storage devices, (flash disc, CD, external disk), software protocols	
Facilitation techniques	Assimilation, Group discussions, Trainer guided • The trainer asks the students to discuss on the paper size of the master layout and present the expert view • The trainer illustrates on the software how to create master layout book, choose & adjust the size and insert the title block with complete relevant information and finally name the master lay • Trainer demonstrate to the students to create master layout book and asks the students to follow the procedures. • The trainer demonstrate to the students how to drag and place the 2D & 3D architectural drawings to the master layout and asks the student to follow the procedures. • The trainer demonstrate to the students how to save these projects in portable document format (PDF) or any other formats compatible to ARCHICAD software and asks the student to follow the procedure.	

	• The trainer tasks each and every student to save their respective project architectural drawings in in portable document format (PDF) or any other formats compatible to ARCHICAD software, this is done by trainee's as he/she monitors. • The trainer uses the saved architectural drawings saved in (PDF) or any other formats compatible to ARCHICAD software to review and further elaborates on how to adjust printing settings in accordance to the size of choice.
Formative assessment methods	• Written assessment • Oral assessment • Practical assessment

Integrated/Summative assessment (For specific module)

Integrated situation

ABC Ltd company has won a contract to conduct an architectural study of a Two (2) storied classroom building in the Nyanza district. The contract scope is about producing and availing all architectural drawings of the building for future construction and permit request processes. Each floor is composed of eight (8) office rooms, one (1) medium hall, four toilet rooms, and one (1) teacher's room. As a designer and employee of the company, you are requested to conduct this study and produce all required architectural drawings in a period not later than three (5) hours. The final drawings will be orally presented after their submission.

Consider the following information:

- ✓ All stories are similar
- ✓ The playground is from the side of the building.
- ✓ The teacher's room is at the entrance of the first floor
- ✓ The building will have a staircase.
- ✓ The toilets are located on the other side with natural ventilation

Resources:

Tools	Mouse and mouse mat, Electrical cables, - Storage devices, (flash disc, CD, External disk), pencils
Equipment	Computer, Printer
Materials/ Consumables	Papers, printer ink

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	

Learning outcome 1: Prepare ArchiCAD software environment	ArchiCAD software is properly installed based on the Operating System	installation file is installed			2
		ArchiCAD license is Activated			2
		ArchiCAD settings are Configured			2
	Software is properly customized as per the installation user guide	Work Environment is Settled Up			2
		Project Templates Configured			2
		Library is Managed			2
		shortcut keys are Used			2
Learning outcome 2: Produce	Project units are properly set up according to the scale of the drawing	working units are Chosen			4
		scale is Chosen			3
		Working area are respected			2

ArchiCAD Drawings	Software tools are Appropriately used according to the design objectives	Design Tools are applied			4
		Documentation Tools are applied			4
		Visualization Tools are applied			2
		Modeling Tools are applied			2
		Collaboration Tools used			2
		Visualization & Rendering tools are applied			4
		Analytical Tools are applied			2
	Software 2D drawings are appropriately created according to the desired design.	Plan views are designed			4
		Section views are designed			4
		Elevation views are designed			4
	Software 3D modeling is appropriately created according to the desired design	3D Navigation are applied			4
		3D rendering are applied			4
	Title block is appropriately created according to the standard requirements	Layout book is selected			4
		New master layout is created			3
		Paper sizes are defined			3
		Title blocks elements are drawn			2
Learning Outcome 3: Print out ArchiCAD drawings	Master layout book is appropriately created according to the scale of the drawing	Paper size and scales are properly adjusted			3
		Title block is allocated			3
	Views are correctly saved and placed on the layout in accordance with to scale of a drawing	2D Views are Dragged and placed			4
		3D Views are Dragged and placed			2
		View settings are Adjusted			2
	Project layout is appropriately saved according to the given project	Master layout is Select			2
		file menu is Selected			2
		file formats are Chosen			2
		PDF drawings are exported			2

	Layout books are appropriately printed out according to the required document format.	Drawings are Printed			2
		Hard copy drawings are Checked			2
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

References

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